Mixer

Category 1

I am a liquid mixer: I mix (for payment, of course) liquids that are delivered to me. I only mix in whole quantities, in whole ratios, and unless otherwise specified, as little as possible. I receive an order, for example:

```
mix A and B in a ratio of 2 to 3; I want a mixture of at least 11 units
```

and then I email the client: 'send me 6 units of A and 9 units of B'. That gives 15 units of the mixture for that customer, but less is not possible if all restrictions must be respected. You also noticed: the client supplies the liquids. I don't even need to know what A or B are … and sometimes I'd rather not know.

That seems simple, but sometimes a customer already sends me some of A and B, hoping it will be enough, for example:

```
mix A and B in a ratio of 2 to 3
with already 4 A and 3 B
i want at least 5 units of mixture
```

What the customer sends me of A and B, I must certainly use. So I consult my spreadsheet and ask the customer for 3 more units of B: that then gives 4 units A and 6 units B, which neatly maintains a 2-3 ratio, is minimal, and keeps everything whole. I hate decimal numbers with non-zeros after the decimal point. It doesn't happen often, but it can be that I don't need to ask the client for anything additional. I inform them of that as well.

That seems simple, but ... I mix any number of different liquids, in arbitrary (whole!) ratios, I want only whole quantities delivered and to process … my spreadsheet needs an upgrade!

Input

The input begins with the number of test cases. Then for each test case:

• One line with the number V of different liquids; 1 < V < 101.

• One line with the ratio of the requested mixture: V positive integers; never zero.

• One line with the minimum size of the order: always greater than zero.

• One line with what the client delivers: for each liquid, a positive integer; this can be zero.

Example Input

4

2

2 3

```
11

0 0

2

2 3

5

4 3

10

1 2 3 4 5 6 7 8 9 10

2060

38 76 114 152 190 228 266 304 342 380

10

1 2 3 4 5 6 7 8 9 10

2110

38 76 114 152 190 228 266 304 342 380
```

Output

Your output must contain, for each test case, one line consisting of first the case number (starting at 1) and a space, then V numbers (each separated by one space) indicating how much of each liquid must still be supplied.

Example Output

```
1 6 9

2 0 3

3 0 0 0 0 0 0 0 0 0 0

4 1 2 3 4 5 6 7 8 9 10
```